

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Artificial Intelligence
2. **Subject Name:** Database Management Systems Lab
L-T-P: 0-0-3 **Credits:** 2
3. **Pre-requisites:** Nil

Syllabus and reference books:

Syllabus:	
Database Design	Entity Relationship Model and Relational Database Design
Structured Query Language	Tables, Databases and Basic queries
	Operators, Clauses and Join operations
	Keys and Indexes
	Advanced operations and Functions
NoSQL Database with MongoDB	Creating Databases and Collections, Basic operations
	Database partitioning, Sharding and Query processing
	Aggregations, Indexing and Searching
Graph database	RDF and SPARQL query language
	Creating, indexing and storing graph database with Neo4J
	Cypher Query language and querying over property graph database
	Predictive analysis on graph database in Neo4J
Reference Books:	
1. Database System Concepts, Deep Learning on Graphs, Abraham Silberschatz, Henry Korth, and S. Sudarshan, McGraw Hill, 6th Edition, 2013	
2. Fundamentals of Database Systems, R. Elmasri and S. Navathe, 7th Edition, Pearson, 2017	
3. MongoDB: The Definitive Guide, Shannon Bradshaw, Eoin Brazil & Kristina Chodorow, 3rd Edition, O'Reilly, 2019	
4. Graph Databases, Ian Robinson, Jim Webber & Emil Eifrem, 2nd Edition, O'Reilly, 2015	
5. RDF Database Systems, Olivier Cure and Guillaume Blin, 1st Edition, Morgan and Kaufmann, 2015	

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Database Design	6
2.	Structured Query Language	12
3.	NoSQL Database with MongoDB	9

4.	Graph database	9
Total number of hours		36